

Design of Questionnaire

-- is both an art and a science

—it's about asking the right questions in the right way to get meaningful, reliable data.

Step-by-Step Guide to Questionnaire Design

1. Define the Objective

- What is to be learnt or measured?
- Frame clear research questions or hypotheses.

2. Identify Target Respondents

- Who will be answering? (e.g., customers, students, professionals)
- Consider their language: literacy level, and familiarity with the topic.

3. Choose the Question Types

Type	Description	Use Case Example
Closed-ended	Multiple choice, Likert scale, Yes/No	Quantitative analysis
Open-ended	Free text responses	Qualitative insights
Rating/Ranking	Rate or rank items	Preferences, satisfaction
Demographic	Age, gender, location, etc.	Segmenting respondents

4. Craft Clear and Neutral Questions

- Keep questions concise and jargon-free.
- Avoid leading or biased wording.

Example: Instead of “How amazing was our service?”

→ “How satisfied were you with our service?”

5. Sequence and Flow

- Start with easy, engaging questions.
- Group similar topics together.
- End with optional open-ended questions (e.g., remark, opinions).

6. Pilot Test the Questionnaire

- Run a small test with a sample group.
- Check for clarity, timing, and technical issues.

7. Finalize and Format

- Use consistent formatting (fonts, spacing).
- Include front matter, and instructions
- Ensure anonymity/confidentiality if needed.

Validity in the context of a questionnaire refers to how well the instrument measures what it is intended to measure. In other words, does the questionnaire actually capture the concept or construct being studied?

Types of Validity in Questionnaires

Type	What It Assesses	Example
Content Validity	Whether the questionnaire fully covers all aspects of the concept. e.g. A survey on job satisfaction includes items on pay, work environment, etc.	
Face Validity	Whether the questions appear relevant and appropriate at face value.	
Construct Validity	Whether the questionnaire truly measures the theoretical construct . Scale measuring "anxiety" correlates with "stress".	
Convergent & Discriminant Validity	Subtypes of construct validity—do related constructs correlate (convergent) and	

Type	What It Assesses	Example
	unrelated ones not (discriminant)?	A scale measuring creativity correlates with innovation but not with aggression.
Criterion Validity	Whether the questionnaire correlates with an external benchmark or outcome	

How to Ensure Validity in Practice

- **Define constructs clearly** before writing questions.
- **Consult experts** to review items for content and face validity.
- **Pilot test** questionnaire and collect feedback.
- Use **Exploratory Factor Analysis (EFA)** or **Confirmatory Factor Analysis (CFA)** to test construct validity.
- Compare results with **external measures** for criterion validity.

Testing the validity of a questionnaire is all about ensuring that instrument truly measures what it's supposed to measure.

1. Content Validity

- Create a **blueprint** mapping each question to specific dimensions of the construct.
- Get **expert reviews** from subject matter specialists.
- Use a **Content Validity Index (CVI)**: Experts rate each item for relevance; calculate proportion of agreement.

2. Face Validity

Does the questionnaire appear valid to respondents and

- Share the questionnaire with a small group of target users.

- Ask for feedback on clarity, relevance, and appropriateness.
- This is subjective but useful for refining wording and tone.

3. Construct Validity

Does the questionnaire measure the theoretical construct?

- Check **convergent validity** (correlates with similar constructs) and **discriminant validity** (does not correlate with unrelated constructs).

4. Criterion Validity

Does the questionnaire correlate with an external benchmark?

- **Concurrent validity:** Compare with an established instrument measuring the same construct.
- **Predictive validity:** See if scores predict future outcomes (e.g., performance, behaviour).

Reliability in a questionnaire refers to its **consistency**—whether it produces stable and repeatable results under consistent conditions. If you administer the same questionnaire to similar respondents at different times (or in different formats), a reliable instrument will yield similar outcomes.

Types of Reliability in Questionnaires

Type	What It Measures	Example
Internal Consistency	How well items within a scale measure the same construct	All items in a “stress” scale should correlate well with each other
Test-Retest Reliability	Stability of responses over time	Administer the same questionnaire twice to the same group, weeks apart
Inter-Rater Reliability	Agreement between different raters or observers	Two reviewers score open-ended responses similarly
Split-Half Reliability	Consistency between two halves of the questionnaire	Compare scores from odd vs. even items

How to Test Reliability

- **Cronbach’s Alpha** (for internal consistency):
 - Values range from 0 to 1

- ≥ 0.7 is generally acceptable
- **Pearson Correlation** (for test-retest or split-half):
 - Measures linear relationship between two sets of scores
- **Cohen's Kappa** (for inter-rater reliability):
 - Adjusts for chance agreement between raters

Cronbach's Alpha (α) is a statistic used to measure the **internal consistency** or **reliability** of a set of items in a questionnaire. In simpler terms, it tells you how well a group of questions work together to measure the same underlying concept.

What It Really Means

- If questionnaire has multiple items intended to assess a single construct (like motivation, anxiety, or satisfaction), Cronbach's Alpha checks whether those items are **correlated** and **consistent**.
- It's expressed as a value between **0 and 1**:

$\alpha \geq 0.9 \rightarrow$ Excellent	$0.8 \leq \alpha < 0.9 \rightarrow$ Good
$0.7 \leq \alpha < 0.8 \rightarrow$ Acceptable	$0.6 \leq \alpha < 0.7 \rightarrow$ Questionable
◦ $\alpha < 0.6 \rightarrow$ Poor	

Formula (for the curious)

$$\alpha = \frac{N \cdot \bar{c}}{\bar{v} + (N - 1) \cdot \bar{c}}$$

Where:

- (N) = number of items
- $(\bar{c})$ = average covariance between item pairs
- $(\bar{v})$ = average variance of each item

Cronbach's Alpha calculation in Excel

Let's say we have a 4-item questionnaire measuring "User Satisfaction" and responses from 10 participants. Set up your Excel sheet like this:

Respondent Q1 Q2 Q3 Q4

1	3	4	3	5
2	2	3	2	4
3	4	5	4	5
4	3	3	3	4

Respondent Q1 Q2 Q3 Q4

5	2	2	2	3
6	4	4	4	5
7	3	4	3	4
8	2	3	2	3
9	4	5	4	5
10	3	4	3	4

Step 2: Calculate Item Variances

Use =VAR.S(range) for each question column:

- Q1: =VAR.S(B2:B11) = 0.6667
- Q2: =VAR.S(C2:C11) = 0.9
- Q3: =VAR.S(D2:D11) = 0.6667
- Q4: =VAR.S(E2:E11) = 0.6222

Sum of item variances = 2.8555

Step 3: Calculate Total Score per Respondent

Add a new column "Total":

=SUM(B2:E2)

Drag down for all rows.

Step 4: Calculate Variance of Total Scores

Use: =VAR.S(F2:F11) Let's say the result is: 10.3222

Step 5: Apply Cronbach's Alpha Formula

$$\alpha = \frac{N}{N-1} \left(1 - \frac{\sum \text{Item Variances}}{\text{Total Score Variance}} \right)$$

In Excel:

= (4 / (4 - 1)) * (1 - (2.8555 / 10.3222))

Result: $\alpha \approx 0.9645$

Interpretation

- If $\alpha \geq 0.7 \rightarrow$ Acceptable reliability
- If $\alpha \geq 0.8 \rightarrow$ Good
- If $\alpha \geq 0.9 \rightarrow$ Excellent

Inter-rater reliability becomes especially relevant when a questionnaire involves **subjective judgments**—where multiple evaluators interpret responses based on their own understanding.

Example: Evaluating Open-Ended Responses in a Job Interview Questionnaire

Imagine you're designing a questionnaire for hiring customer service representatives. One section includes open-ended questions like:

“Describe a time you handled a difficult customer.”

Now, three HR professionals independently rate each response on a scale of 1 to 5 for:

- Empathy
- Problem-solving
- Communication clarity

Here's where **inter-rater reliability** matters:

- If Rater A gives a response a 5, Rater B gives it a 2, and Rater C gives it a 4, the inconsistency suggests poor inter-rater reliability.
- But if all three raters consistently score similar responses similarly, it shows the rating system is **stable and objective**.

Why It's Important

- Ensures fairness in scoring
- Validates the consistency of your measurement tool
- Helps refine rating criteria or training for raters

You could use **Cohen's Kappa** or **Intraclass Correlation Coefficient (ICC)** to quantify this reliability, depending on the scale and number of raters.